

TEST-TYPES

§11.4.169 Test-Type Coverage — vasic.digital + milosvasic.ru

This repository ships **two static sites** (vasic.digital, a standalone OpenDesign build; milosvasic.ru, a Jekyll build) plus **Go/Node/Python tooling** that generates and validates them. There is **no application server, no database, and no runtime state** — the deployment target is GitHub Pages (static file hosting). That fact determines which of the §11.4.169 mandatory test types are *applicable*, which are *added by this change*, and which are *genuinely N/A* (documented here per §11.4.3 SKIP-with-reason rather than left as a silent gap).

Status legend: **PRESENT** (existed before this change) · **ADDED** (this change) · **N/A** (not applicable to a static-site + build-tooling repo, with cited reason).

#	§11.4.169 test type	Status	Where / Evidence
1	unit	ADDED	Go table tests for the generator: <code>_tools/gen/*_test.go</code> (markdown renderer, <code>esc/escText</code> , frontmatter parser, <code>availableLangs</code> , <code>langPath</code> , <code>Site.URL</code> , hreflang matrix, JSON-LD nodes, <code>seoHead</code> singletons, <code>sitemap/robots</code> writer, <code>license-hide</code> , <code>chip-cap</code> , <code>repoLabel</code>). Python unit tests for the glossary sentinel roundtrip: <code>_tests/tools/glossary_protect_test.py</code> . Evidence: <code>evidence/test-types/go-test.txt</code> ,

#	§11.4.169 test type	Status	Where / Evidence
2	integration	ADDED	evidence/test-types/glossary-protect.txt. Link-integrity crawl across each generated site — nav, product cards, portfolio, downloads, and every sitemap.xml URL must resolve <400; the anti-bluff private-repo deep-link guard is re-asserted. _tests/tests/link-integrity.spec.js. Also the data-gate mutation test proving validate.mjs actually fails on bad portfolio data (excluded / private-leak / order-break): _tests/tools/validate.mutation.test.mjs → evidence/test-types/validate-mutation.txt. Per key page: no inline secrets in shipped HTML, no http:// mixed-content subresources, no inline on*= handlers, rel=noopener on every target=_blank, and CSP-<meta> presence recorded. _tests/tests/security-hardening.spec.js. Server-header security (CSP/HSTS response headers, WAF) is N/A — GitHub Pages does not let you set response headers; the only static-site lever is a <meta> CSP, which is checked. Real transferred-bytes + request-count budget measured from Playwright's network layer (not a faked Lighthouse score), plus LCP-candidate presence
3	security	ADDED	
4	performance	ADDED	

#	§11.4.169 test type	Status	Where / Evidence
5	e2e	PRESENT	and a render-blocking-<head>-script check. Lighthouse/LHCI is not a declared/installable dependency here, so a documented lighter proxy is used per the task allowance. _tests/tests/perf-budget.spec.js → real numbers in evidence/test-types/perf-budget.json (e.g. vasic home 221 KB / 10 req / LCP 28 ms; budget 1.5 MB / 45 req). Feature/behaviour specs: _tests/tests/vasic-digital.spec.js, milosvasic-ru.spec.js, *-features.spec.js (nav, hero, theme toggle + persistence, product-card links, responsive overflow).
6	accessibility (a11y)	PRESENT	@axe-core/playwright audit + keyboard/skip-link/contrast checks: _tests/tests/milosvasic-ru-ally.spec.js.
7	i18n	PRESENT	_tests/tests/i18n-completeness.spec.js, vasic-digital-i18n-articles.spec.js, milosvasic-ru-i18n-articles.spec.js; hreflang reciprocity is additionally asserted by the new seo-meta.spec.js.
8	visual	PRESENT	Visual oracle + evidence screenshots: _tests/visual/visual-oracle.js, _tests/visual/self-validate.sh, per-site full-page screenshots under evidence/homepages/.
9	export	PRESENT	PDF/asset export validation: _tests/export/validate-pdf.js, make-

#	§11.4.169 test type	Status	Where / Evidence
10	SEO / structured-data	ADDED	image.js, self-validate.sh, fixtures. Per key page: exactly one non-empty <title>, non-empty meta description, canonical, valid Open Graph + Twitter card, ≥1 parseable JSON-LD block with a schema.org @type, and reciprocal hreflang. _tests/tests/seo-meta.spec.js. (Folded under §11.4.169 SEO coverage.)
11	ddos / rate-limiting	N/A	No origin server to flood. GitHub Pages is a CDN-fronted static host; there is no application request handler, session, or compute path this repo owns that could be rate-limited or DoS-tested. Any DDoS protection is a property of GitHub's edge, not of this codebase. Nothing to assert without bluffing.
12	concurrency / atomicity	N/A	The shipped artifact is immutable static HTML/CSS/JS with no shared mutable state, no transactions, no writes at request time. The only concurrency in the repo is the offline build (gen, translate pipeline), which is single-shot and deterministic; its correctness is covered by the unit tests above (pure functions, no goroutine-shared state). There is no concurrent-write invariant to test.
13	memory / leak	N/A	No long-running process owned by this repo. Pages are served as files; the client-side JS is a tiny

#	§11.4.169 test type	Status	Where / Evidence
14	chaos / resilience	N/A	deferred motion/theme controller with no allocation loop or retained listeners across navigations (each page is a fresh document). The Go generator is a short-lived CLI that exits; there is no resident heap to profile for growth. (Client memory pressure would surface via the perf-budget asset ceiling.) Chaos engineering targets distributed runtime failure modes (node loss, network partition, dependency brownout). A static site has no backend topology, no inter-service calls, and no failover to perturb. The analogous “resilience” property for a static site — that no page links to something that 404s or leaks a private repo — is covered as integration (link-integrity + private-repo guard) rather than chaos.

Notes on honest edges (no bluff)

- **Private-repo suppression lives at the data gate, not in the Go renderer.** `_tools/gen` declares `PortfolioEntry.Private` but never branches on it — the renderer trusts pre-validated data. Suppression is enforced upstream by `_tools/portfolio/validate.mjs` (fails if a `private:true` entry lists repos) and re-asserted at runtime by the Playwright private-repo deep-link guard. The Go side adds `TestPortfolioDataHasNoPrivateLeak` as a belt-and-suspenders guard on the data the renderer consumes. This is documented rather than papered over.
- **milosvasic.ru homepage has no hreflang.** Its `<head>` is owned by `jeekyll-seo-tag`, which does not emit `hreflang`. Its product and portfolio pages ARE Go-rendered and carry the

full reciprocal hreflang matrix, as does all of vasic.digital. seo-meta.spec.js therefore asserts hreflang on every page EXCEPT that one homepage; the exception is intentional and recorded here, not silently skipped.

- **Performance uses a measured proxy, not Lighthouse.** No Lighthouse score is reported because LHCI is not an installed dependency. The budget is real measured transfer size / request count / LCP presence — see evidence/test-types/perf-budget.json for the actual per-page numbers.

How to run

```
# unit – Go generator
( cd _tools/gen && go test ./... )

# unit – glossary sentinel roundtrip
python3 _tests/tools/glossary_protect_test.py

# integration – data-gate mutation proof
node _tests/tools/validate.mutation.test.mjs

# integration / security / performance / SEO / e2e / ally / i18n
# LOCAL specs only, 3 browsers – the default config `testIgnore`s
the three
# live-production specs below, so this is 19 of the 22 spec files.
( cd _tests && npx playwright test )

# LIVE production specs (restyle-seo-regression, v170-fixes, v171-
hardcoding,
# all-languages-link-integrity) – chromium only, asserts against
the deployed
# sites over the public internet, so it needs working DNS/egress.
The
# all-language crawl additionally needs VD_BASE/MV_BASE, which
default to
# localhost:8401/8082 and are set by _tools/deploy-langs.sh at
deploy time;
# without them that one spec fails with "Expected: 200 / Received:
0".
( cd _tests && npx playwright test --
config=playwright.live.config.js )
```